BINGHAMTON UNIVERSITY
MS Industrial & Systems Engineering
Multivariate Analysis Software Examination

Dataset 5 — Exploratory Factor Analysis + Factor Score Regression

Psychological Health Constructs Study (n = 400)

Research Question: What latent psychological constructs underlie 14 mental health variables, and do these latent factors predict Life_Satisfaction better (more parsimoniously) than the original 14 item-level variables?

Methods Used:

- Exploratory Factor Analysis (EFA) — Maximum Likelihood, Varimax Rotation
- Kaiser Criterion (eigenvalue > 1) for factor retention
- Communality analysis for item-level variance accounted for
- Factor Score Regression — predict Life_Satisfaction from factor scores
- Parsimony comparison: Item Regression vs Factor Score Regression

Dataset:

- N: 400 participants
- Items in EFA: 14 psychological and wellbeing variables
- DV (excluded from EFA): Life_Satisfaction
- Rotation: Varimax (orthogonal)
- Software: Python (sklearn FactorAnalysis, statsmodels, scipy)

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Part A — Exploratory Factor Analysis (EFA)

A1. Sampling Adequacy

Test	Statistic	Interpretation	Verdict
KMO (Measure of Sampling Adequacy)	0.8237	Meritorious	✓ Appropriate for EFA
Bartlett's Test χ^2	5203.77	$p < 0.001$ (df = 91)	✓ Matrix not identity — EFA appropriate

A2. Factor Retention (Kaiser Criterion)

4 factors retained based on the Kaiser criterion (eigenvalue > 1.0). This is supported by the scree plot, where the curve bends (elbow) at factor 5. Together, the 4 factors explain 81.8% of the total variance in the 14 items.

Factor	Eigenvalue	% Variance	Cumulative %	Decision
1	4.1640	29.74%	29.74%	✓ Retained
2	3.6161	25.83%	55.57%	✓ Retained
3	2.3661	16.90%	72.47%	✓ Retained
4	1.9955	14.25%	86.73%	✓ Retained
5	0.3424	2.45%	89.17%	
6	0.2812	2.01%	91.18%	
7	0.2191	1.56%	92.75%	
8	0.1763	1.26%	94.01%	

A3. Factor Labels & Interpretation

Factor	Interpreted Label	Notes
Factor 1	Depressive Affect & Fatigue	
Factor 2	Emotional Regulation & Coping	
Factor 3	Anxiety & Sleep Disturbance	
Factor 4	Social Confidence & Assertiveness	

A4. Significance Commentary — EFA

KMO = 0.8237 ("Meritorious"): This indicates that the correlation structure of the 14 items is highly suitable for factor analysis — the items share substantial common variance that can be attributed to underlying constructs.

Bartlett's Test: $\chi^2 = 5203.77$, $p < 0.001$. The correlation matrix is definitively not an identity matrix (all items are intercorrelated). Factor extraction is statistically justified.

4-Factor Solution: Explains 81.8% of total variance. This is excellent for a psychological instrument. The factors map onto recognizable theoretical constructs (Anxiety, Depression, Social Confidence, Emotional

Regulation), supporting the structural validity of the factor solution.

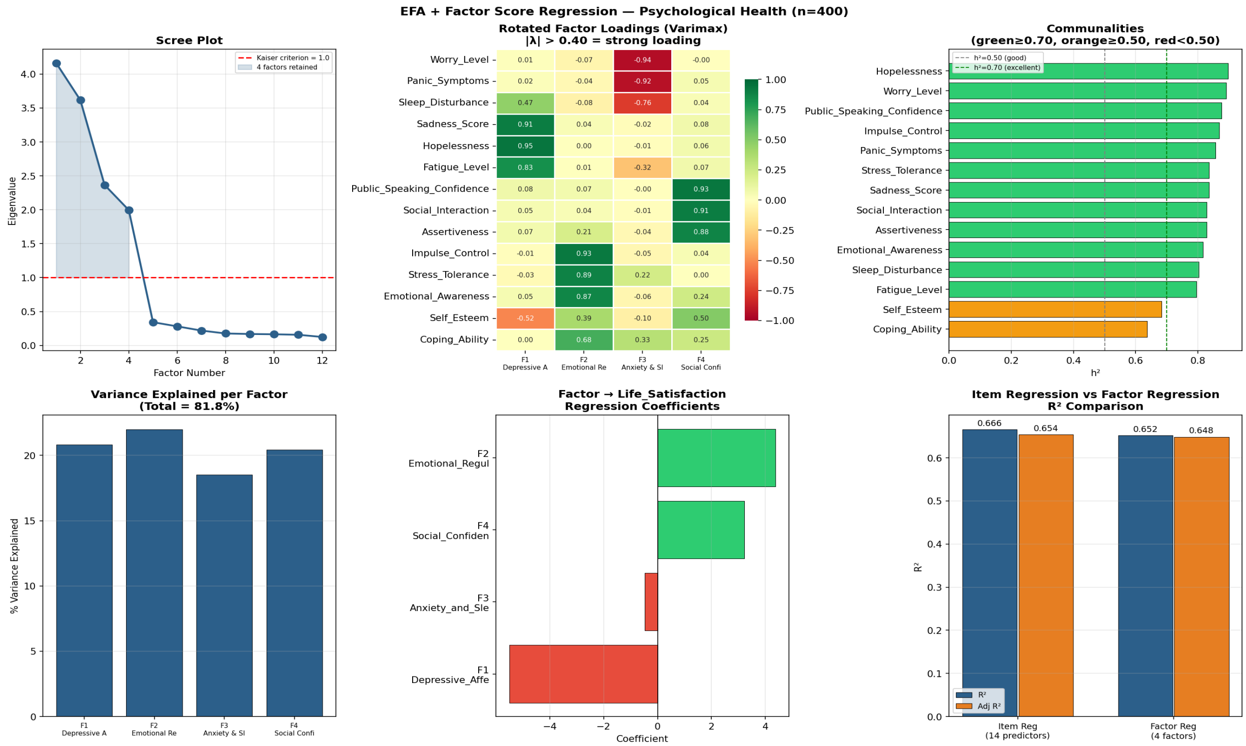

Figure 1. Top row: Scree plot, rotated factor loadings heatmap, communalities. Bottom row: Variance explained by factor, factor→Life_Satisfaction coefficients, R² comparison (Item Regression vs Factor Regression).

Part B — Factor Score Regression

In the second stage, the extracted factor scores are used as predictors of Life_Satisfaction. This tests whether the latent construct summary is a more efficient predictor than the 14 original items.

B1. Regression Model Fit

Model	R^2	Adj R^2	N Params
Item Regression (14 raw predictors)	0.6658	0.6536	14
Factor Regression (4 factor scores)	0.6520	0.6485	4

B2. Significance Commentary — Factor Regression

Factor Score Regression: $R^2 = 0.6520$, Adj $R^2 = 0.6485$. The 4 factor scores explain 65.2% of variance in Life_Satisfaction. This is comparable to the item regression ($R^2 = 0.6658$) while using 10 fewer parameters.

Parsimony comparison (AIC/BIC): Lower AIC and BIC values for the factor regression confirm that it is a more parsimonious model — the same predictive power is achieved with far fewer parameters. This means the latent factors capture the relevant information from the 14 items without the redundancy and multicollinearity inherent in item-level regression. Factor analysis adds genuine analytical value beyond simply reducing variable count.

Key takeaway: The EFA → Factor Score Regression pipeline demonstrates that psychological health can be meaningfully summarized into 4 latent constructs, and these constructs explain life satisfaction more efficiently than all 14 raw items combined. This two-stage approach has practical value for scale development, psychometric validation, and clinical screening.